Labour Market Integration of Immigrants in Canada Post-Pandemic (2021)

A Predictive and Descriptive Analysis

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Introduction

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This is Lisa.

This is Amy.

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Lisa is 28 years old & has a Bachelor of Science in Data Analytics.

Amy is 27 years old & has a Bachelor of Science in Data Analytics.

They both have 5 years of experience.

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If two people within the same age range, had the same education credentials, would you assume they had the same chances of getting hired?

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Not necessarily.

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Today's job market is tough for everyone.

But do immigrants face additional barriers…

especially after COVID?

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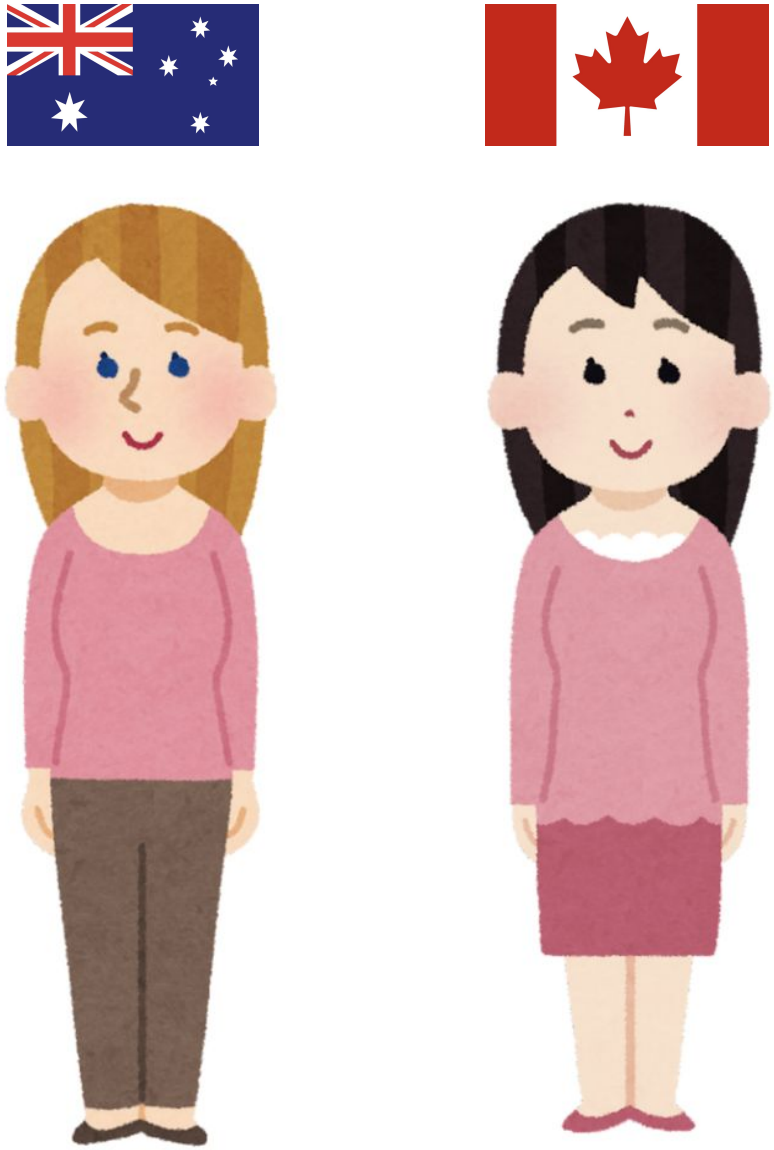
Immigrant Employment Rate: ~90.6%



Non-Immigrant Employment Rate: ~92.8%

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Forbes

Subscribe: Less than $1.50/wk

What Does A Worker Want? What The Labor Shortage Really Tells Us

By Tom Spiggle, Former Contributor. ⓘ I'm an employment lawyer who...

Published Jul 08, 2021 at 12:33pm EDT, Updated Dec 1

CBC

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Business

Labour shortage hampering post-pandemic recovery for businesses in Canada, study finds

55% of entrepreneurs can't find enough workers, leading to delays, limiting growth

Menu

x · CBC News · Posted: Sep 29, 2021 6:00 AM EDT | Last Updated: September 30, 2021

POLICY OPTIONS POLITIQUES

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Post-COVID labour market disruptions created uneven employment conditions.

gration

Business

Where Canada's job market stands one year after the pandemic began

The country added 259,000 jobs in February, but there are still 599,000 fewer people working than a year ago

CBC News · Posted: Mar 12, 2021 8:41 AM EST | Last Updated: March 12, 2021

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Canada selects immigrants based on **skills** and **education**. Yet employment gaps still exist.

But, why?

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Research Questions

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Do age and education explain employment outcomes?

Do differences persist between immigrants and non-immigrants?

How well can we predict employment using these factors?

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Data & Approach

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Data: 2021 Census (Statistics Canada)

Aggregated dataset (group-level data)

Key variables:

- Age
- Education
- Immigrant status

Two Approaches:

1. Predictive Modelling
2. Descriptive Analysis

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Methodology

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Models Used:

- Logistic Regression
- Random Forest

Model Setup:

- Train/Test Split (80/20)

Evaluation:

- Accuracy
- Precision & Recall
- F1 Score

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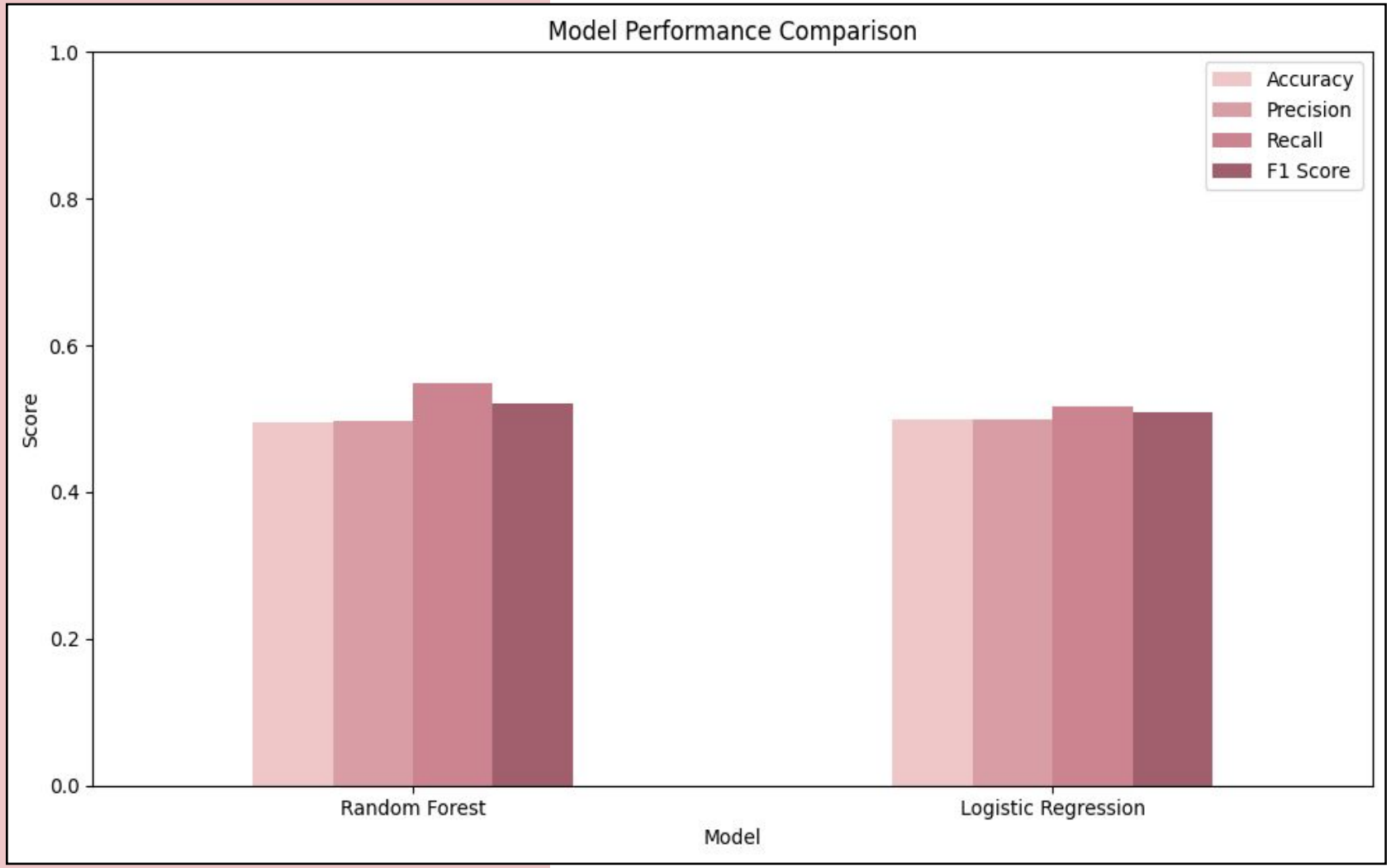

Accuracy: 50%

F1 Score: 0.50

Models perform weakly.

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Model Results

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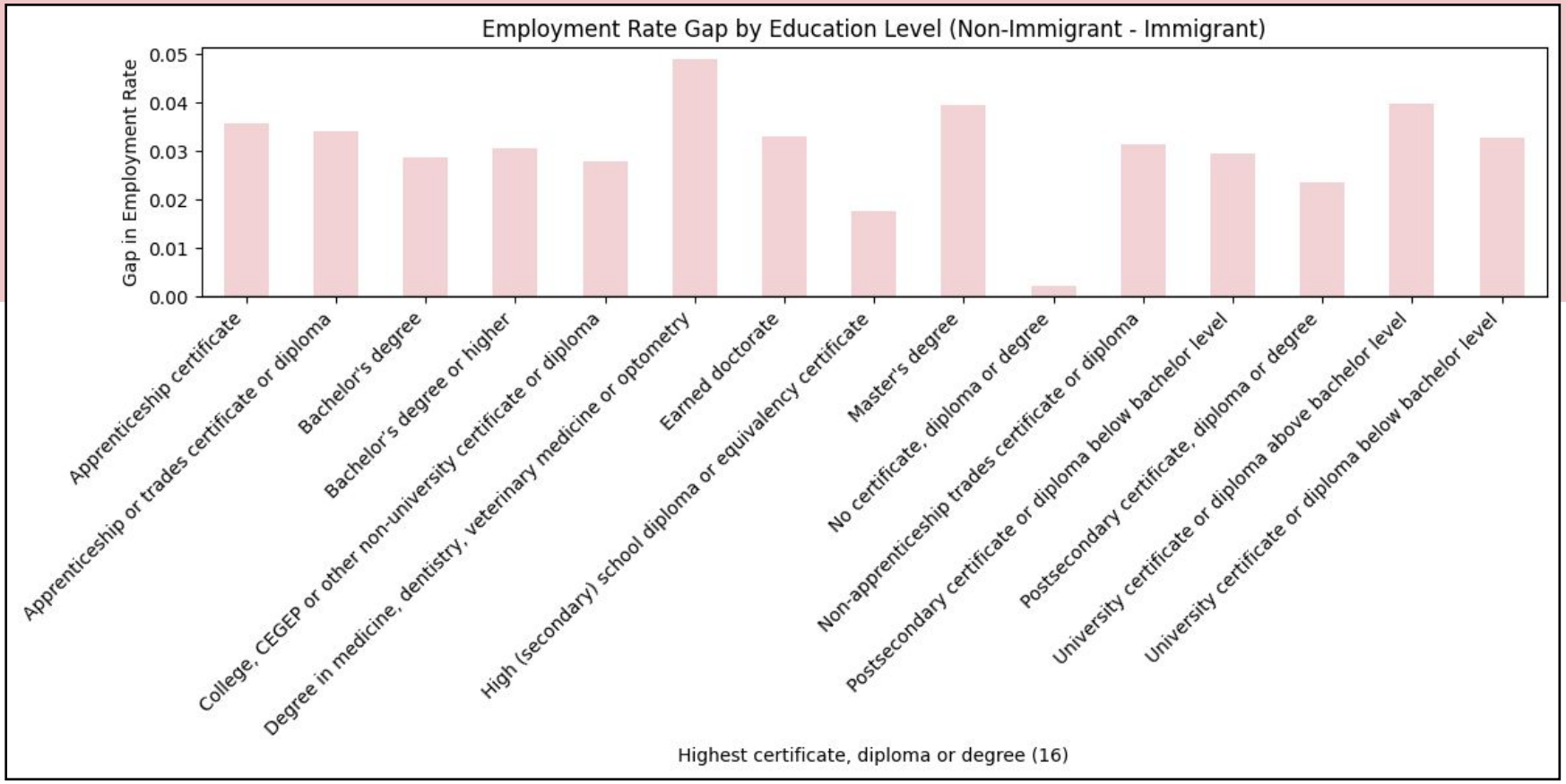
Largest gaps in specialized fields

Smallest gaps in lower-skilled roles

Higher education → higher employment ?



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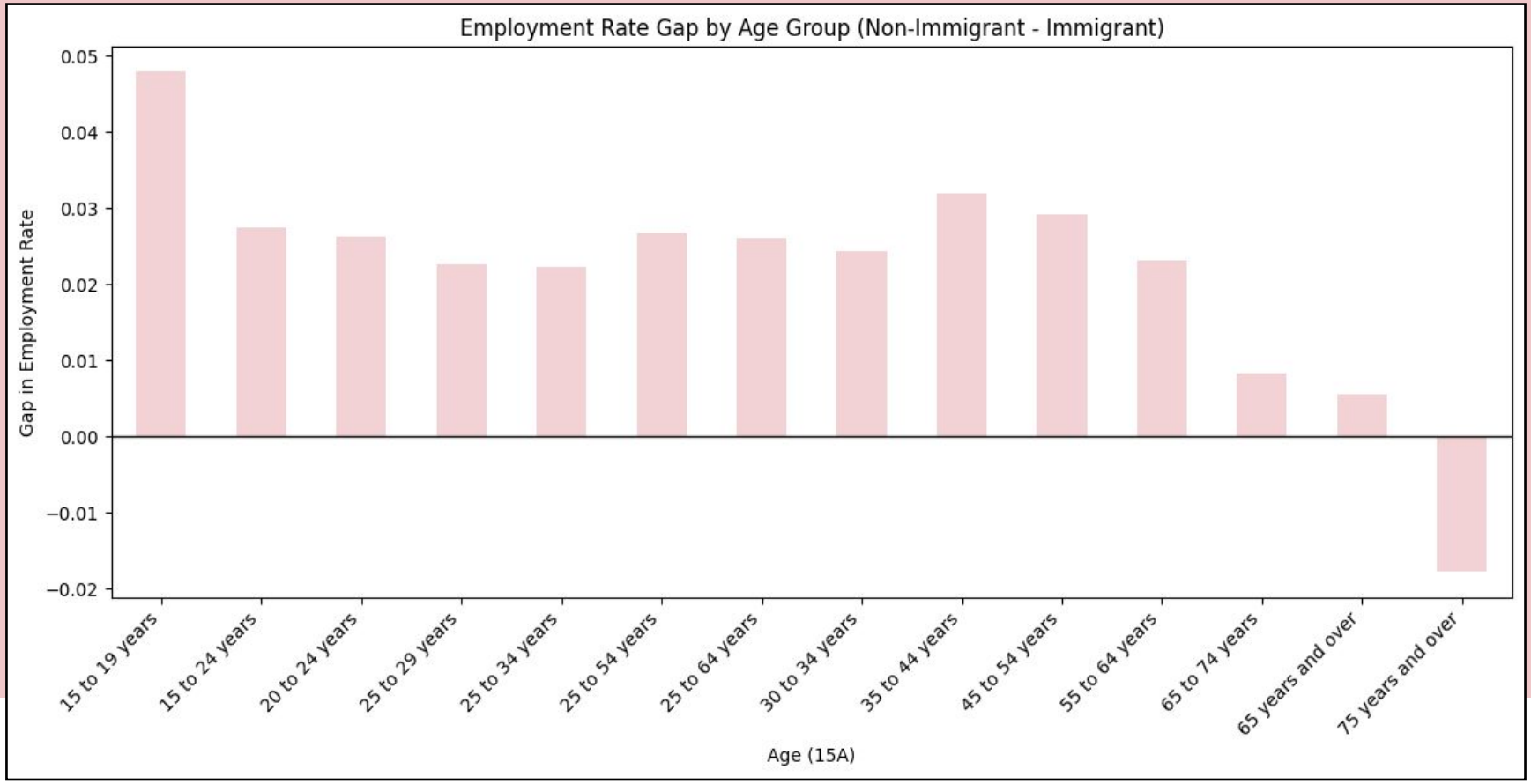

Lower Employment Rate for younger immigrants

Higher Employment Rate for older immigrants

Gradual labour market integration ?

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Age:

- younger immigrants: language barriers, education prioritization, lack of experience
- older immigrants: lack of savings, benefits

Education:

- low skills: matching credentials, lower necessity for language skills
- high skills: regulated fields, certifications

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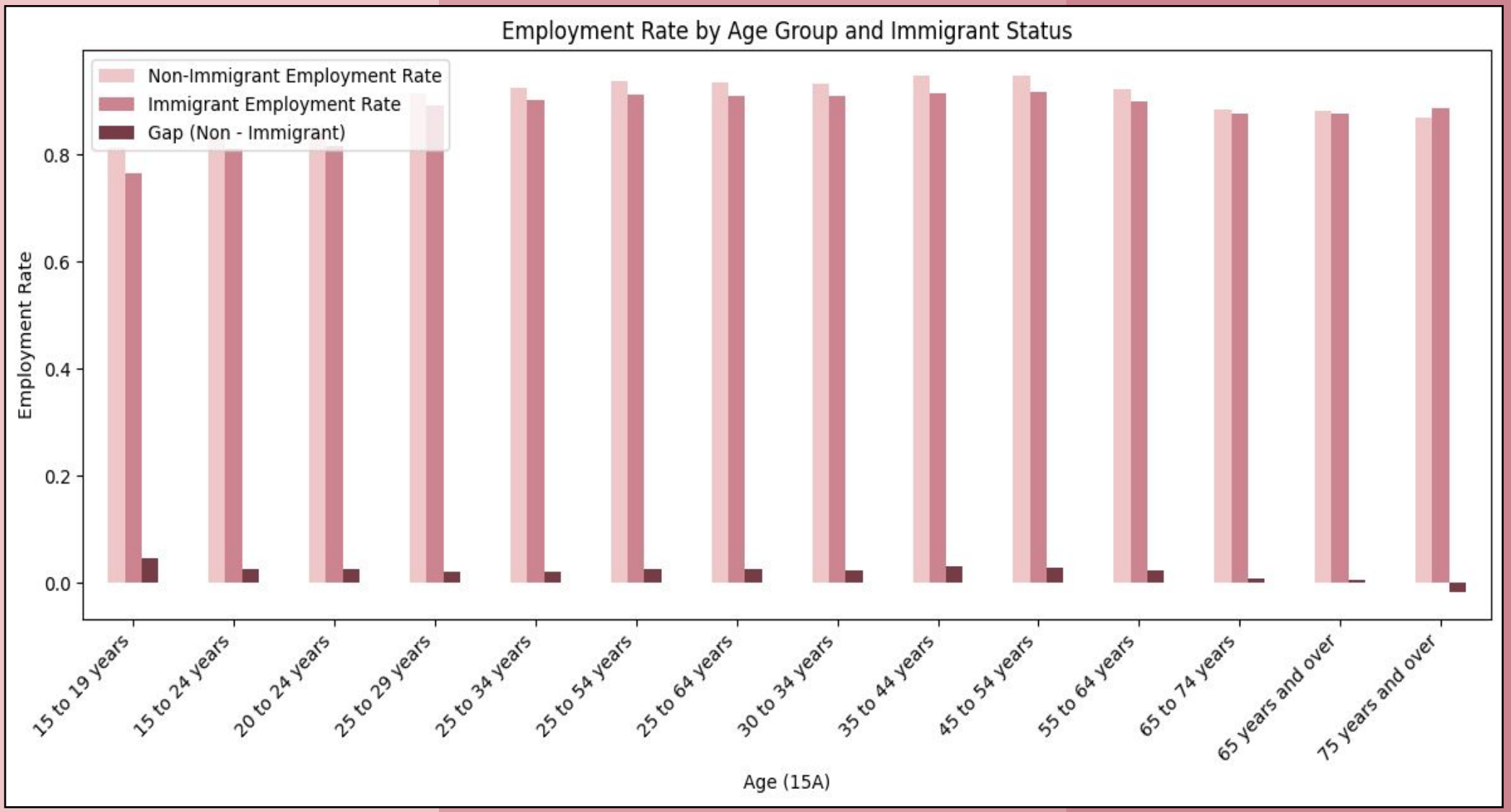

Less than 0.05% difference

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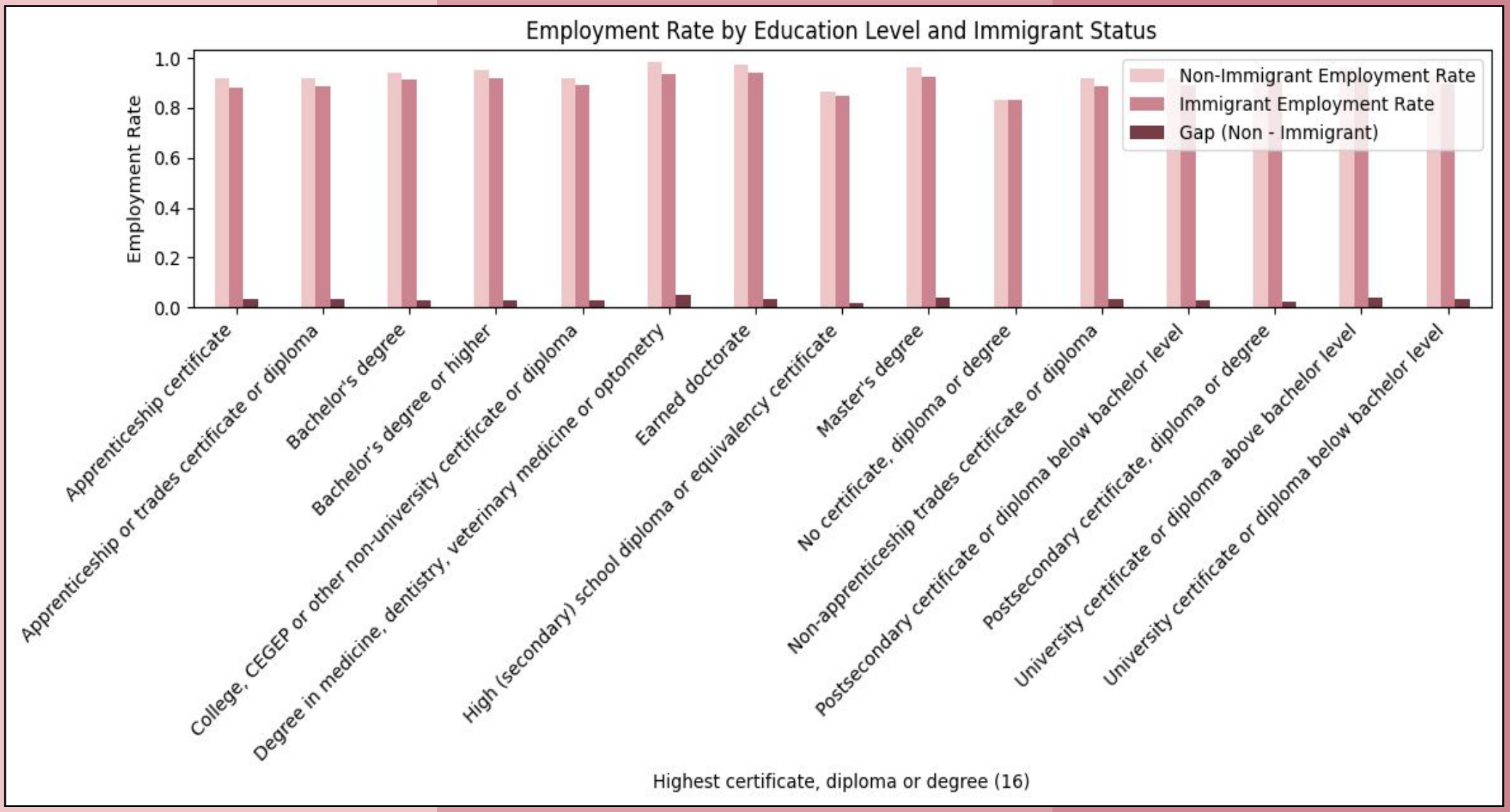

Less than 0.05% difference

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Age and education DO NOT strongly predict employment outcomes!

So then what patterns do we actually see across different groups?

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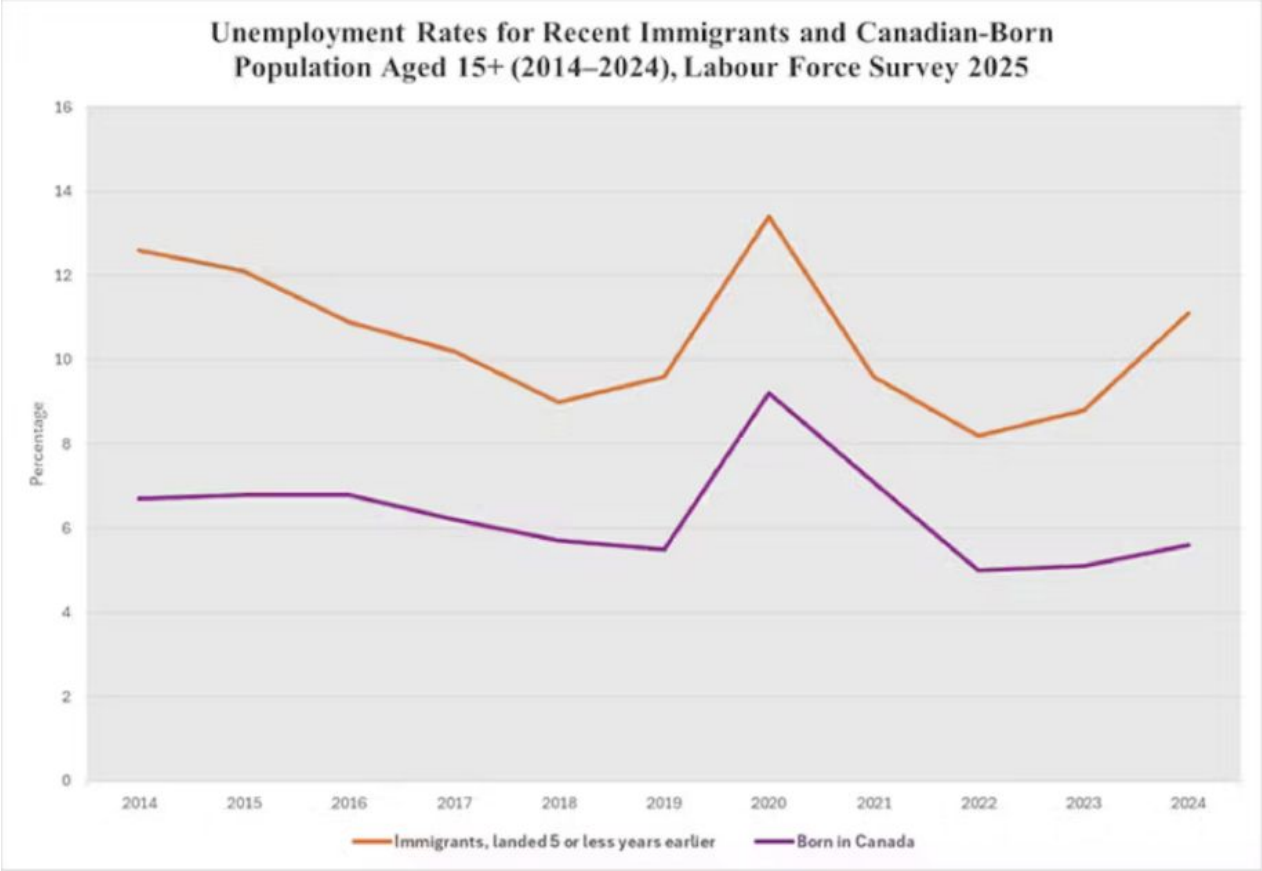

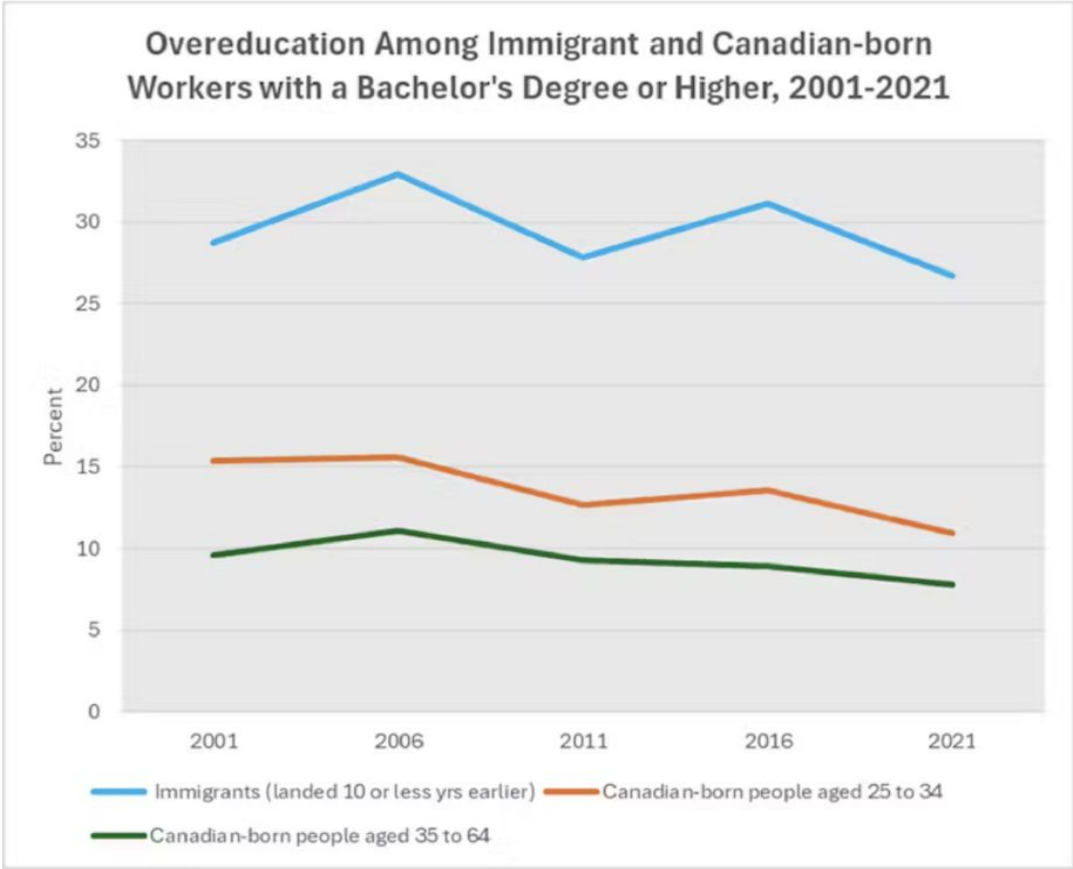

Underemployment is also a persistent issue for immigrants. In 2021, only 44 per cent of immigrants who had arrived in Canada within the previous decade were employed in jobs matching their education level, compared to 64 per cent of Canadian-born workers aged 25 to 34.

The over-education rate — the proportion of university graduates working in jobs for which they are over-qualified despite holding a bachelor's degree or higher — was 26.7 per cent for immigrants, more than double the 10.9 per cent rate for Canadian-born workers in 2021.

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Limitations

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Data:

- Aggregated data (group-level)
- Missing variables (language, experience)
- Single time point (post-COVID snapshot)
- No causal conclusions

Employment status is a simplified outcome

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Future Work

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Analyze multiple Census years (long-term trends)

Integrate Statistics Canada Labour Force Survey data

Examine job quality and underemployment

Explore industry and occupation differences

Conclusion

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- **Age & Education matter, but do not explain employment gaps.**
- **Structural factors play a key role.**
- **Understanding these factors is essential for developing more *effective and equitable* labour market policies.**

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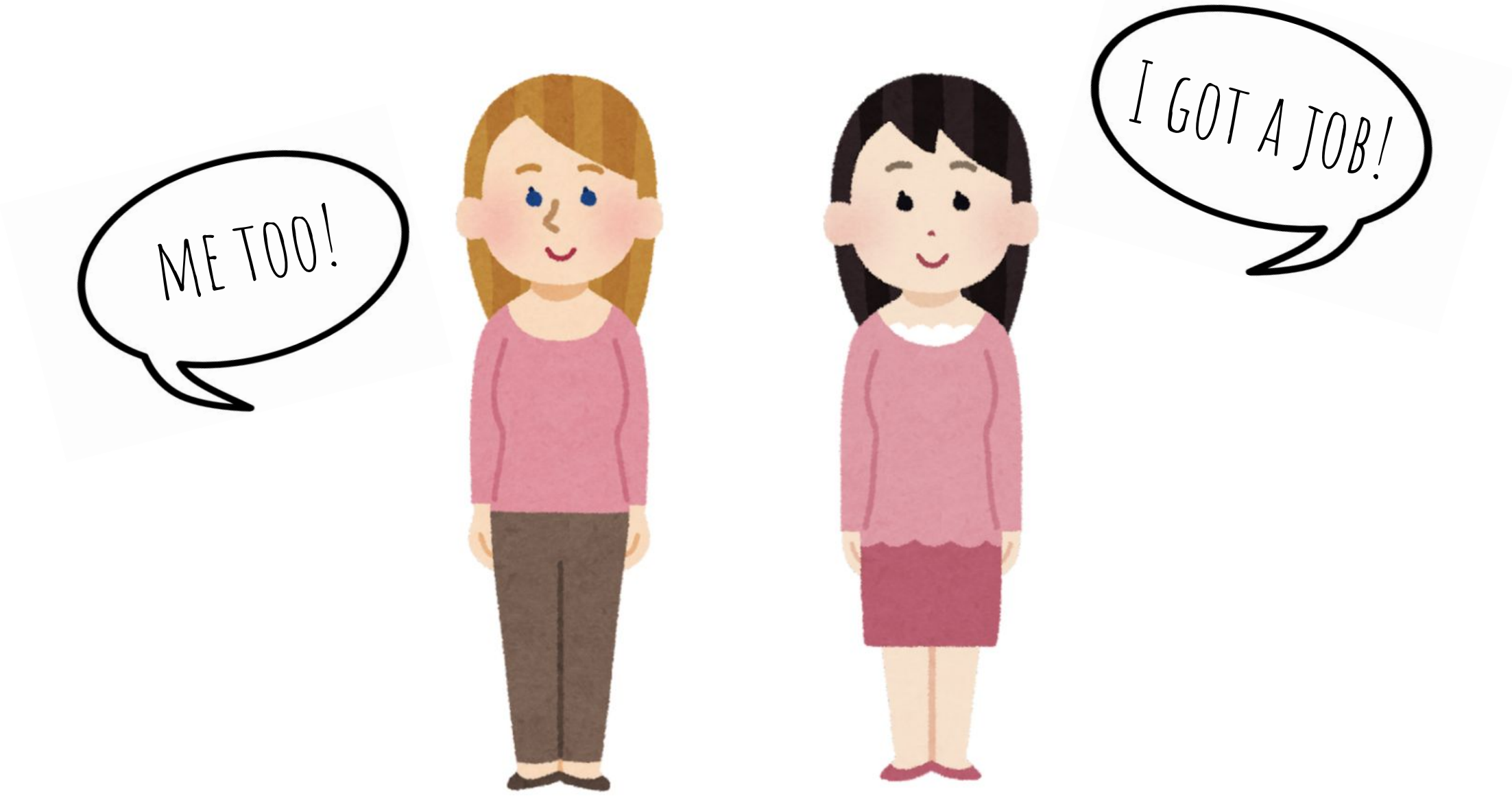

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Thanks for listening!

Q&A

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